Dynamic Behavior of Microbial Populations in Stirred Bioreactors Simulated with Euler−Lagrange Methods: Traveling along the Lifelines of Single Cells†

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In the present work, an Euler−Lagrange approach has been applied to characterize the behavior of a heterogeneous cell population in a stirred-tank bioreactor with nonideal mixing. It allows one to describe population behavior as the outcome of the interaction between the intracellular state of its individual cells and the turbulent flow field in the reactor. The modeling approach and the numerical method employed are based on an Euler−Lagrange formulation of the system combined with a fractional-step method to allow for a stable, accurate, and numerically efficient solution of the underlying equations. This strategy permits one to account for the heterogeneity present in real reactors in both the fluid and cellular phases, respectively. Three examples are used to illustrate the application of this approach. The suggested method is easily transferable to similar problems requiring a segregated and high-dimensional description of the internal structure of the corpuscular phase, which usually precludes the application of population balance equations.

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Introduction

The physiological state of cellular systems and its impact on growth and product formation is the result of a complex interplay between the extracellular environment and the cellular machinery. The functionality of a biosystem for the purpose of bioproduction processes is therefore determined by the cooperative action of extracellular stimuli and the intracellular makeup of the biological phase. The design of bioreactors in which living cells function as factories as well as the prediction of suitable operating conditions is further complicated because of the dynamic variations of the extracellular environment. Consequently, a quantitative description of these phenomena should rest upon the two interlinked aspects of structured bioprocess modeling: The first aspect concerns the complex interaction of the functional units within each cell, including the mathematical formulation of intracellular reaction rates and key regulatory responses of these networks to environmental changes. The second aspect involves the structure of the abiotic phases of the bioreactor for analysis of the impact of spatial and temporal variation in the intensities of mixing and mass transfer, leading to concentration gradients of various substrates and products. A modeling framework suited for capturing local variations in both intra- and extracellular concentrations should therefore rest on a link between metabolic network modeling and computational fluid dynamics (CFD).

Previous attempts addressing this issue have almost been exclusively based on the Euler−Euler approach in which gas, liquid, and biophase are considered as a continuum.[1−4] However, microorganisms are cellular in nature, and the continuum description is not rigorously correct. In fact, the continuum approach leads to a loss of realism if the individual history of the cells becomes the focus of attention, e.g., when considering cumulative starvation effects in cells during fed batch fermentations. In some instances, these problems can be tackled by combining the Euler approach for the fluid phases with population balance equations (PBEs).[5] An inherent limitation of the PBE approach, however, is that the incorporation of a detailed intracellular reaction network leads to a computationally intractable model already for ideally mixed systems because a high-dimensional distribution function must be computed.[6] The complementary approach of combining CFD simulations with population balances based on metabolically unstructured models of single cells faces similar practical limitations.[5] To overcome this problem, it is sometimes possible to resort to hybrid approaches combining multizonal models with CFD calculations, as reported by Bauer and Eigenberger[7,8] for the case of gas−liquid bubble columns. Recently, Bezzo et al.[5] applied this strategy to xanthan gum production in stirred tanks. The authors combined an Eulerian description of the fluid flow with a multizonal model in which the reactor was divided into a limited number of spatial regions. These were assumed to be well-mixed and homogeneous and to be capable of material exchange with adjacent zones. Within each of these zones, a mass-structured population balance was formulated that was then combined with an unstructured kinetic model of substrate consumption and xanthan production.

Alternatively, one can account for the intracellular structure of the cells while neglecting the spatial concentration gradients in the liquid phase. This corresponds to a Lagrange-type formulation of the equations for the corpuscular phase in an ideally mixed system.[6,9−11] This description has the obvious limitation that bioprocesses are performed in real reactors where spatial variations in concentration usually cannot be neglected. Detailed mathematical models capturing the variation in both the extracellular environment and the metabolism of the segregated biophase promise to aid significantly in describing the behavior of cell popula-

† Dedicated to Prof. Dr.-Ing. Gerhart Eigenberger on occasion of his 65th birthday.

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tions in bioreactors. This requires a combination of both approaches outlined above.

To incorporate intracellular reactions within a computationally tractable model, the approach presented in this contribution is based on an Euler−Lagrange simulation. Here, modeling of the extracellular environment is still based on the continuous Euler approach, whereas the behavior of the biophase is characterized by a discrete cell-ensemble approach (Lagrange). Similar simulation strategies have been applied to describe the mixing behavior of tracer particles in stirred tanks[12] or the behavior of the gas phase in bubble columns.[13] The cell-ensemble approach implies the assumption that the continuum limit represented by the PBE model can be approximated by a sufficiently large number of individual cells.

The numerical solution of the resulting system is severely limited by the immense computing power required for three-dimensional discretization in space and the necessity to populate each of the resulting finite volumes with a sufficient number of cells to minimize the effect of statistical error on the accuracy of the solution. This problem cannot be overcome with the numerical methods routinely embedded in available software packages. Here, we present an alternative solution strategy that can be applied to this kind of problem. The application of this approach is illustrated with the following examples for three-dimensional turbulent flow in a stirred-tank reactor: (1) the simulation of glucose concentration fields during fed batch cultivation of the yeast *Saccharomyces cerevisiae*, (2) the mixing of a tracer substance, and (3) the temporal and spatial behavior of a population of oscillating yeast cells. The latter example represents the first successful integration of CFD and intracellular metabolism employing a segregated description of the biophase.

Theoretical Basis

Euler−Lagrange Approach. Previous CFD simulations of stirred tanks in bioreaction engineering have almost exclusively employed the Euler−Euler approach, treating both the fluid phases and the cell population as a continuum. The aforementioned work of Bezzo et al.[5] represents a notable exception in this regard. A more realistic description of the biological phase needs to reflect the fact that the state of the cell population is actually dependent on the previous history of its individual cells. The intracellular state of these single cells at a given time t^* is determined by their specific initial state at $t = 0$ and by their particular trajectories in the inhomogeneous fluid phase during the period $0 < t < t^*$. It is possible to capture this variation by using an Euler−Lagrange approach, where a continuum representation of the fluid phases is combined with a segregated description of the cell population. This allows each single−computational−"cell", which still represents a large collective of real cells, to be endowed with its individual intracellular structure and state. In the modeling and simulation approach, mass transfer and reactions are assumed not to affect the turbulent flow field. The three-dimensional turbulent flow for single- and two-phase systems has been simulated with the commercial software package PHOENICS employing a modified Chen−Kim $k-\epsilon$ turbulence model.[3,4,14] The population of microorganisms is distributed over the reactor, which is subdivided into finite volumes. Convective movement of the single cell is computed from

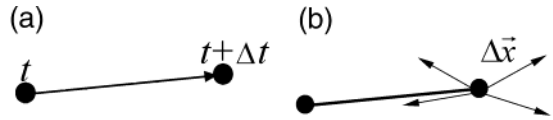


Figure 1. Total particle displacement during a time step between t and $t + \Delta t$ results from the superposition of (a) convective particle movement and (b) random particle movement, $\Delta\vec{x}$, due to turbulent dispersion in the reactor.

the velocity field $\vec{V}$ assuming that no slip occurs between the fluid and the corpuscular phase (cf Figure 1a)

$$\frac{d\vec{x}}{dt} = \vec{V} \tag{1}$$

where $\vec{x}$ represents the position vector of the individual cells. Slip can be neglected because the convective fluid velocity can be shown to surpass the slip velocity of microbial cells by several orders of magnitude. Momentum transfer between the particles and the fluid phase also does not require explicit consideration because the suspension can be treated as a quasi-single phase for particles smaller than the mesh spacing, as shown previously.[13]

Within a finite volume, the velocities of the liquid phase in each spatial coordinate are linearly approximated, which allows one to obtain an analytical solution of eq 1 inside the finite volume. Thus, a numerical solution of eq 1 is unneeded.

The random movement caused by the turbulent dispersion is superimposed on the convective flow (cf Figure 1b)

$$\Delta x_i = \xi_i\sqrt{2D_T\Delta t} \tag{2}$$

where Δx_i is the random jump in one dimension (x, y, or z) during a time step of Δt, ξ_i signifies a random number with a variance of 1, and $D_T = f(k,\epsilon)$ stands for the local eddy diffusivity calculated from the CFD simulation as a function of the turbulent kinetic energy k and the turbulent energy dissipation rate ϵ.

Prediction of the intracellular state of a single cell along the trajectory is performed by incorporating the system of intracellular balance equations into the model

$$\frac{d\vec{c}_{in,m}}{dt} = \mathbf{A}_m\vec{r}_m(\vec{c}_{in,m}(t),\vec{c}_{ex}(\vec{x},t)) \tag{3}$$

Here, $\vec{c}_{in,m}$ denotes the vector of intracellular metabolite concentrations in the individual cell m and $\vec{c}_{ex}$ is the concentration vector of extracellular compounds at the position of this cell ($\vec{x}$). $\mathbf{A}_m$ stands for the stoichiometric matrix of the metabolic network in cell m, and the term $\vec{r}_m$ represents its vector of intracellular reactions rates, which, in general, are nonlinear functions of $\vec{c}_{in,m}$ and $\vec{c}_{ex}$.

Inclusion of $\vec{c}_{ex}$ in the intracellular balance equations is required for the description of transport processes across the cell membrane, i.e., substrate uptake and product excretion. These occur in the cellular reaction rates $\vec{r}_m$ and additionally are considered as a source term, $\vec{S}$ (sink/source), in the Euler simulation of the extracellular state. Assuming that T of a total of R intracellular reaction rates represent transport rates across the cell membrane, the system can be written as

$$\frac{\partial\vec{c}_{ex}(\vec{x})}{\partial t} + (\vec{V}\,\nabla)\vec{c}_{ex}(\vec{x}) = \mathrm{div}(D_T\,\mathrm{grad}\vec{c}_{ex}(\vec{x})) + \vec{S}(\vec{x}) \tag{4}$$

with

$$\vec{S}(\vec{x}) = \frac{\varphi V_R}{N_C}\sum_{m=1}^{N_C} \mathbf{A}_{T,m}\vec{r}_{T,m}(\vec{c}_{in}(t),\vec{c}_{ex}(\vec{x},t))\,\delta(\vec{x}-\vec{x}_m), \quad T \le R \quad (5)$$

where $\vec{S}(\vec{x})$ equals the vector of net transport rates of all metabolites across the cellular membrane of all cells present at position $\vec{x}$. This is obtained by multiplying every addend of the sum over all N_C cells by Dirac's delta function, which signifies whether cell m (with position $\vec{x}_m$) is present at $\vec{x}$ or not. The subvector $\vec{r}_{T,m}$ of $\vec{r}_m$ contains only those T reaction rates accounting for transport across the cell membrane. $\mathbf{A}_{T,m}$ represents the associated submatrix of $\mathbf{A}_m$ consisting of columns that correspond to the transport reactions included in $\vec{r}_{T,m}$. The term preceding the sum reflects the influence of the total cell volume on the intensity of the exchange: φ denotes the ratio of the whole cellular volume $N_C V_m$ and the reactor volume V_R, where V_m corresponds to the volume of a (simulated) single cell. Expression 4 accounts for convection and turbulent diffusion in the liquid phase and describes the coupling between the extracellular environment and the intracellular metabolism of the single cells.

Numerical Methods. The numerical solution of the problem posed by the set of eqs 1−4 is challenging for several reasons: (i) The set of intracellular balance equations (eq 3) and the exchange term accounting for transport across the cellular membrane $\vec{S}(\vec{x})$ in eq 4, result in a highly nonlinear system. (ii) The resulting system is stiff because of the general properties of intracellular balance equations (eq 3) and also because of the diffusion term in eq 4. Simulation runs demonstrate that the system of eqs 1−4 cannot be solved with an explicit integration algorithm with reasonable accuracy in most cases of practical relevance. (iii) The system of equations resulting from discretization is of enormous size due to the facts that (a) the problem is of three-dimensional nature and (b) a sufficiently large number of cells (typically around 10^5) is required to achieve a realistic description of the population.

In the following discussion, each of the challenges mentioned above will be addressed in detail.

The size of the total set of equations can be calculated from

$$N_{eq} = N_{ex}M^3 + (N_C N_{st}) + 6N_C \quad (6)$$

with N_{eq} representing the total number of equations, N_{ex} being the number of extracellular state variables, M^3 denoting the number of grid nodes, N_C representing the total number of microbial cells, and N_{st} signifying the number of intracellular state variables of each single cell. The term $6N_C$ derives from the separate expressions for convection and diffusion of the microorganisms in three spatial dimensions. Thus, the achievable spatial resolution is very limited already for laboratory-scale reactors, typically on the order of 10^5 computational nodes (e.g., $50^3 = 125\,000$ nodes for a grid with 50 discrete points in all three coordinates) are considered. This results in comparatively large grid Peclet numbers ($Pe_G = V\Delta x/D_T$), even for small stirred tanks, on the order of $1 < Pe_G < 20$. The numerical difficulties arising from the choice of the spatial discretization grid represent a general problem in accurately solving material balances,[15] and they are especially critical for scale-up. Regarding the large stirred-tank volumes generally encountered in industrial biotechnology (up to 500 m^3), it is therefore desirable to develop computationally efficient strategies that allow for maximization of the spatial and temporal grid spacing while permitting an accurate solution of the material balances. The method presented below has the potential to substantially reduce the computational effort required to solve this kind of system and, thus, to make the method fit for numerical scale-up.

Spatial interpolation within the small control volumes of the discretized solution domain is based on a third-order Agarwal interpolation method with a total variation diminishing (TVD) limiter, thus minimizing numerical dispersion and the problem of nonphysical oscillations encountered with most traditional methods.[15−17]

Apart from the numerical difficulties arising from the spatial discretization, integration in the time domain also proves to be demanding. A straightforward approach to solve the system of eqs 1−4 would involve treating all terms simultaneously, replacing each of the derivatives by finite differences, and advancing in time. However, the presence of a diffusion term in eq 4 and the system of intracellular balances in eq 3 result in a stiff system necessitating the application of an implicit integration algorithm providing at least second-order accuracy to avoid severe time step restrictions.

This second-order accuracy could easily be obtained using a standard Crank−Nicolson method for time discretization and central differences in space. However, such an approach would lead to a method in which all terms are implicit, with all components exhibiting nonlinear coupling at every grid node. The resulting enormous set of nonlinear algebraic equations would need to be solved simultaneously, which would require a great computational effort.

By contrast, the computational effort can be drastically reduced by employing an operator splitting method. This approach is based on a transfer of the well-known splitting or approximate factorization approach for linear systems to our nonlinear problem.[15,17−19] Integration of the resulting split system over time is then performed separately for each part with an appropriate method providing second-order accuracy. An adaptive step size control has been employed to determine a suitable time step for the full system. Thus, the computation time required for the integration of the large system under consideration (cf eq 6) can be remarkably reduced without losing accuracy. This high accuracy is an essential prerequisite for a successful solution of this system because of the numerical problems encountered when applying routine integration methods.

The basic idea of the splitting method is to decompose the system into a sum of terms, F_α, where each single term describes a process that can be calculated independently of the others and where each term must also be of stable nature. This allows for the application of the most suitable integration method to each of the processes. It must be kept in mind, however, that this decomposition method also has limitations: e.g., no general proof exists that it is always possible to find a suitable decomposition for nonlinear systems, and the stability of the solution of the full system cannot be guaranteed.

If such a decomposition is feasible, the discretized set of equations can be formulated as

$$\frac{d\vec{Y}}{dt} = \sum_{\alpha=1}^{N} \vec{F}_\alpha(\vec{Y},t) \tag{7}$$

where $\vec{Y}$ denotes the state vector representing the full system, N is the number of processes into which the system is decomposed, and $\vec{F}_\alpha$ stands for the vector of expressions describing process α.

Numerical integration in time of the discretized equation set by a second-order procedure on the interval $t_{k-1} \leq t \leq t_{k+1}$ can be written as follows[19,20]

$$\begin{aligned}
\frac{\vec{Y}^{\,k+1/N-1} - \vec{Y}^{\,k-1}}{\Delta t} &= F_1\left(\frac{\vec{Y}^{\,k+1/N-1} + \vec{Y}^{\,k-1}}{2}, t_k\right) \\
&\vdots \\
\frac{\vec{Y}^{\,k} - \vec{Y}^{\,k-1/N}}{\Delta t} &= F_N\left(\frac{\vec{Y}^{\,k} + \vec{Y}^{\,k-1/N}}{2}, t_k\right) \\
\frac{\vec{Y}^{\,k+1/N} - \vec{Y}^{\,k}}{\Delta t} &= F_N\left(\frac{\vec{Y}^{\,k+1/N} + \vec{Y}^{\,k}}{2}, t_k\right) \\
&\vdots \\
\frac{\vec{Y}^{\,k+1} - \vec{Y}^{\,k+(N-1)/N}}{\Delta t} &= F_1\left(\frac{\vec{Y}^{\,k+1} + \vec{Y}^{\,k+(N-1)/N}}{2}, t_k\right)
\end{aligned} \tag{8}$$

where the index k denotes the kth time step. Instead of using the above approximation (centered in time) given by eqs 8, it is also possible to employ alternative methods that yield accuracies of second order or higher.

The system under discussion is very amenable to the use of splitting methods given that it can be easily separated into a number of mutually independent processes: (i) $\vec{F}_1$, convective particle movement; (ii) $\vec{F}_2$, random movement of cells due to turbulent dispersion; (iii) $\vec{F}_3$–$\vec{F}_5$, convection and diffusion of extracellular compounds in x, y, and z directions, respectively; (iv) $\vec{F}_6$, source term describing the material exchange between cells and the liquid phase; and (v) $\vec{F}_7$, intracellular balance equations describing cellular metabolism.

Another special advantage of applying a fractional-step method to this problem is that, frequently, the contributions $\vec{F}_\alpha$ are no longer coupled after being split and, therefore, they can be solved in parallel. Three-dimensional convection and diffusion of extracellular compounds can be decomposed into separate terms, $\vec{F}_3$–$\vec{F}_5$, each of which describes only one-dimensional transport in one of the spatial coordinates. Before proceeding with time integration for $\vec{F}_3$–$\vec{F}_7$, the aforementioned high-resolution TVD space schemes are applied to $\vec{F}_3$–$\vec{F}_5$. By contrast, the equation systems $\vec{F}_1$ and $\vec{F}_2$ are solved using alternative methods: $\vec{F}_1$ is calculated from an analytical function obtained from spatial interpolation of the velocity field as described above, and $\vec{F}_2$ is computed using eq 2 for random movement. The application of this strategy is illustrated for several examples in the following section.

Results and Discussion

Growth and product formation of cell populations in stirred-tank bioreactors is strongly dependent on the abiotic environment provided and on the history of the individual cells. CFD simulations of the turbulent flow field in the reactor have been combined with an Euler–Lagrange representation of the system and the numerical strategies outlined above. This allows for both the intrinsic heterogeneity of the cell population (Lagrange) and the concentration gradients present in the stirred tank (Euler) to be captured.

Three simulation examples are discussed in the following subsections to illustrate the difference between the Euler–Euler and Euler–Lagrange descriptions of the bioreactor that will serve to demonstrate the potential of the numerical solution strategy outlined above. All simulation results were visualized employing the visualization software package OPEN INVENTOR.[21]

Example 1: Coupling of Computational Fluid Dynamics with Unstructured Kinetic Models for *Saccharomyces cerevisiae*. The following example demonstrates that the suggested numerical strategy can also be applied to Euler–Euler-type simulations, which is exemplified for baker's yeast production in a fed batch. The Euler–Euler approach combines an unstructured and unsegregated description of the continuous biophase with a three-dimensional CFD simulation for the reactor. The concentration of the carbon and energy source in the feed is as high as possible (in the range of 500 kg m^{-3}). Inside the tank, the substrate concentration very often is in the range of the saturation constant, i.e., in the concentration range of 1–100 g m^{-3}. The challenge in the scale-up of these processes is then to prevent concentration gradients resulting in further limitations ($c_S < c_{S,crit}$), as well as formation of unwanted byproducts such as ethanol if $c_S > c_{S,crit}$.

Assuming that the oxygen supply is sufficient to avoid local oxygen limitations, the kinetic model required for the simulation includes only the material balance equations for the substrate and the product ethanol. As suggested by Bajpai and Reuss,[22] the uptake kinetics is considered only in the vicinity of the so-called critical sugar concentration. Thus, a rather simple unstructured empirical model is chosen for the purpose of this study. The sink term describing the local consumption of the substrate glucose for biomass growth and formation of the byproduct ethanol then reads

$$\begin{aligned}
S_S(\vec{x}) = &-\frac{1}{Y_{X/S}}\mu^{max}\frac{c_S(\vec{x})}{K_S + c_S(\vec{x})}c_X(\vec{x}) - \\
&\frac{1}{Y_{P/S}}\nu^{max}\frac{\max(0, c_S(\vec{x}) - c_{S,crit})}{K_P + \max(0, c_S(\vec{x}) - c_{S,crit})}c_X(\vec{x})
\end{aligned} \tag{9}$$

where $S_S(\vec{x})$ denotes the local substrate uptake rate. Biomass growth is described by monod kinetics, and the corresponding substrate consumption is determined via the yield coefficient $Y_{X/S}$. μ^{max} represents the maximum specific growth rate; K_S is the half-saturation constant with respect to the substrate; and $c_S(\vec{x})$ and $c_X(\vec{x})$ stand for the concentrations of substrate and biomass at position $\vec{x}$, respectively. Ethanol formation occurring for values of $c_S(\vec{x}) > c_{S,crit}$ is modeled in a similar fashion with the maximum specific ethanol production rate ν_{max} and the half-saturation constant K_P. The resultant glucose consumption for this process is determined by the yield coefficient $Y_{P/S}$. Because ethanol production is modeled as being solely dependent on the extracellular glucose concentration, only eq 9 is used as a source term in the material balance equation

$$\frac{\partial c_S(\vec{x})}{\partial t} + \frac{\partial[u_j(\vec{x})\, c_S(\vec{x})]}{\partial x_j} = \frac{\partial}{\partial x_j}\left[D_T\frac{\partial c_S(\vec{x})}{\partial x_j}\right] + S_S(\vec{x}) \tag{10}$$

The three-dimensional velocity field $\vec{u}(\vec{x})$ of the tank and the turbulent dispersion coefficient D_T are available from CFD simulations assuming a turbulent Schmidt

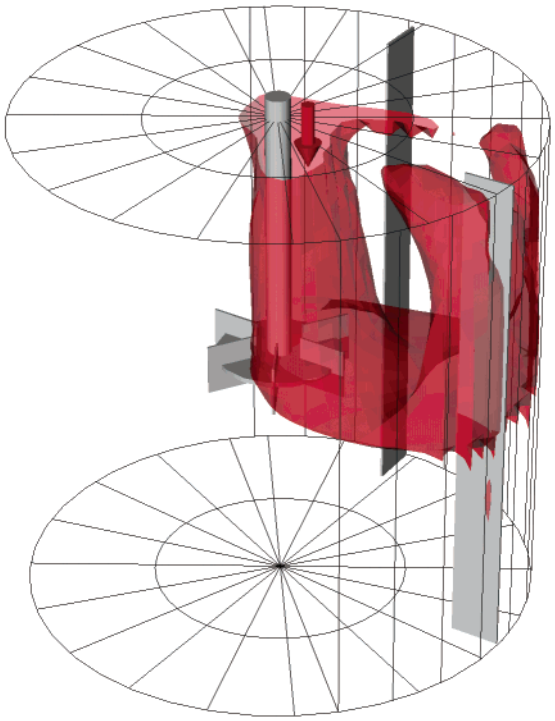


Figure 2. Three-dimensional simulation of the stationary substrate distribution during a yeast fed-batch fermentation in a 68-L laboratory-scale stirred tank equipped with a Rushton turbine. Red indicates regions of ethanol production ($c_S > c_{S,crit}$). The simulation was performed for the vessel geometry given by Costes and Couderc[23] (tank height and diameter = 0.44 m, height of center of impeller blade above ground = 0.22 m, impeller diameter = 0.147 m, four baffles of width 0.044 m) for an impeller speed of N_{imp} = 165 rpm. The arrow signifies the point of feed addition.

number of $Sc = 1$.[4] Additionally, the substrate feed rate (Fc_S^0) is considered as a source term in the volume element corresponding to the feeding point, where F denotes the feed flow rate and c_S^0 is the sugar concentration in the feed medium. The resulting system of balance equations was solved by employing the numerical methods described above, but with a continuum description of the biophase instead of the Lagrange approach. Figure 2 shows the stationary substrate distribution when glucose is fed at the top of a 68-L laboratory fermentor equipped with a Rushton turbine. The red-shaded region indicates glucose concentrations leading to ethanol excretion due to overflow metabolism (Crabtree effect). Asymmetric accumulation of glucose is caused by eccentric position of the concentrated glucose feed. The corresponding situation for a three-impeller system in a larger stirred tank (900 L) is illustrated in Figure 3.

In addition to the cloud of high glucose concentrations leading to ethanol formation at the top of the tank (yellow), the second envelope between the first and second impellers borders the fraction of the volume with glucose concentrations below the critical value for maintenance metabolism (blue). Frequent entrance of yeast cells into this region can cause serious starvation effects. The pronounced gradients in the tank reflect the poor axial pumping and thus mixing capabilities of Rushton turbines.

Example 2: Simulation of a Mixing Experiment Employing a Tracer Substance. To compare the accuracy of three different numerical approaches for a given spatial resolution ($M^3 = 44^3$), the mixing behavior of a tracer substance in a 68-L laboratory-scale stirred tank[23] was calculated. The simulation results are displayed in Figure 4. Obviously, when the Euler approach is used for the particle phase, the simulation performed with the UPWIND method is not able to reproduce the multiple peaks that are observed in real tracer signals. By contrast, the TVD algorithm yields a significantly more realistic description of the tracer behavior in this case. When the Lagrange strategy is applied with 100 000 tracer particles, good agreement with the Euler−TVD simulation is observed. This demonstrates that the behavior of this system can be described well with the chosen resolution and number of particles.

Figure 3. Three-dimensional simulation of glucose distribution during a yeast fed batch fermentation in a three-impeller system (N_{imp} = 400 rpm). The two-fluid simulation (gas/liquid) was performed for a 1500-L vessel filled with 900 L of liquid (tank height = 1.76 m, tank diameter = 0.83 m, impeller diameter = 0.33 m, height of center of lowest impeller blade above ground and space between impellers = 0.46 m, four baffles of width 0.064 m). The arrow signifies the point of feed addition. Yellow-shaded regions indicate ethanol production in the presence of high glucose concentrations due to the Crabtree effect ($c_S > c_{S,crit}$), whereas blue volumes mark glucose concentrations below the critical value for maintenance metabolism.

Example 3: Synchronization of Glycolytic Oscillations in a Population of Yeast Cells. This final example illustrates how the Euler−Lagrange approach can be applied to simulate the dynamic interactions between the single cells of a population and their spatially heterogeneous environment in a stirred tank. To this end, the case of yeast glycolytic oscillations has been selected. The model describes the synchronization dynamics of a population of yeast cells exhibiting autonomous glycolytic oscillations that emerge from the

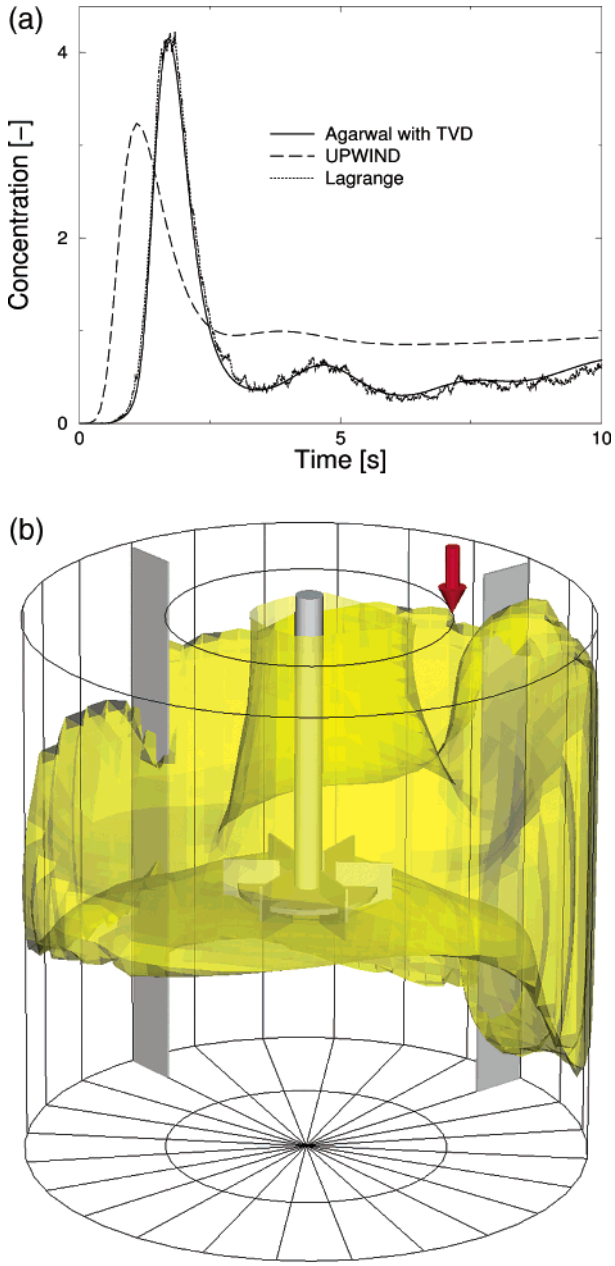
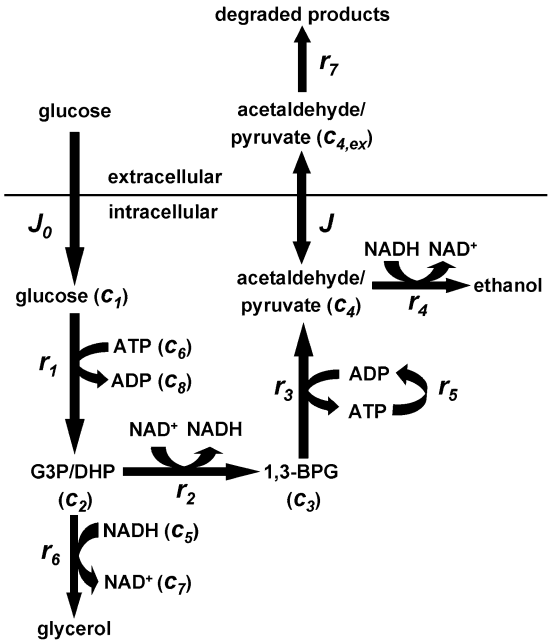


Figure 4. (a) Pulse response in 3D flow at the measurement point after addition of the tracer substance calculated with different numerical methods: Euler approach with UPWIND method (dashed line), Euler approach with Agarwal–TVD scheme (solid line), and Lagrange description (dotted line). Vessel geometry, injection point, and operating conditions correspond to the setup shown in Figure 2. For the Lagrange case, 100 000 tracer particles were added at the injection point at time $t = 0$. The simulated measurement point of the depicted concentration trajectories is located near the wall at the height of the impeller. (b) Three-dimensional visualization of the tracer cloud calculated with the Agarwal–TVD scheme.

interactions of the internal structure with the three-dimensional turbulent flow. The kinetic model employed here (cf Scheme 1) was suggested by Wolf and Heinrich.[24] The single cell model consists of six nonlinear ordinary differential equations (ODEs) comprising glucose transport into the cell, conversion of intracellular glucose into two C_3 molecules including the associated phosphorylation steps, and glycerol formation linked to NADH consumption. For the lower part of the glycolysis, two lumped reactions including NADH and ATP production are considered, as well as ethanol production

Scheme 1. Model of Yeast Glycolysis as Suggested by Wolf and Heinrich[24]



and the excretion or uptake of the lumped intermediate pools of pyruvate and acetaldehyde. The metabolic model is capable of predicting oscillations in glycolytic metabolites at the single-cell level. For a list of kinetic rate equations, parameters, and initial conditions for the single-cell model employed for the simulation of glycolytic oscillations, please see Tables 1–3. The kinetic rate equations (Table 1) as well as most parameter values (Table 2) were obtained from Wolf and Heinrich.[24] Parameter values for J_0 and κ and the set of initial conditions shown in Table 3 were taken from Henson et al.[6] The total number of simulated cells in the reactor, N_C, was used specifically for this study. A more extended discussion of the model can be found in the original paper of Wolf and Heinrich.[24]

Scheme 2 illustrates the set of balance equations for the individual cells of the population. The synchronization of these individual oscillators at the population level is directly related to the net flux of acetaldehyde/pyruvate, J_m, from cell m into the extracellular environment. J_m is modeled as a permeation term where the flux is proportional to the difference between the individual intracellular concentration and the concentration of the common extracellular pool in a finite volume. The mass balance on extracellular acetaldehyde/pyruvate shows that all cells present at position $\vec{x}$ contribute to and interact with this pool (cf eq 5). Employing the terminology of systems theory, this concentration can be therefore be regarded as the attractor for the observed synchronization at the population level (cf Figure 5).

To assess the influence of the mixing intensity on the ability of the yeast population to synchronize their autonomous oscillations, the three-dimensional model based on the Euler–Lagrange approach is simulated for two different turbulent regimes and is compared to the solution obtained for the ideally mixed case presented in Henson et al.[6] To follow the process of synchronization, the population of yeast cells is randomly initialized to yield an asynchronous starting culture. This was achieved by perturbing the initial conditions of the

Table 1. Balance Equations of the Single-Cell Model of Glycolysis According to Wolf and Heinrich[24] (Ideally Mixed Case)[a]

metabolite	balance equation
intracellular glucose	$\frac{dc_{1,m}}{dt} = J_0 - r_{1,m} = J_0 - k_1 c_{1,m} c_{6,m}\left[1 + \left(\frac{c_{6,m}}{K_I}\right)^q\right]^{-1}$
intracellular G3P/DHP	$\frac{dc_{2,m}}{dt} = 2r_{1,m} - r_{2,m} - r_{6,m} = 2k_1 c_{1,m} c_{6,m}\left[1 + \left(\frac{c_{6,m}}{K_I}\right)^q\right]^{-1} - k_2 c_{2,m}(c_{5\mathrm{tot},m} - c_{5,m}) - k_6 c_{2,m} c_{5,m}$
intracellular 1,3-BPG	$\frac{dc_{3,m}}{dt} = r_{2,m} - r_{3,m} = k_2 c_{2,m}(c_{5\mathrm{tot},m} - c_{5,m}) - k_3 c_{3,m}(c_{6\mathrm{tot},m} - c_{6,m})$
intracellular acetaldehyde/pyruvate	$\frac{dc_{4,m}}{dt} = r_{3,m} - r_{4,m} - J_m = k_3 c_{3,m}(c_{6\mathrm{tot},m} - c_{6,m}) - k_4 c_{4,m} c_{5,m} - J_m$
intracellular NADH	$\frac{dc_{5,m}}{dt} = r_{2,m} - r_{4,m} - r_{6,m} = k_2 c_{2,m}(c_{5\mathrm{tot},m} - c_{5,m}) - k_4 c_{4,m} c_{5,m} - k_6 c_{2,m} c_{5,m}$
intracellular ATP	$\frac{dc_{6,m}}{dt} = -2r_{1,m} + 2r_{3,m} - r_{5,m} = -2k_1 c_{1,m} c_{6,m}\left[1 + \left(\frac{c_{6,m}}{K_I}\right)^q\right]^{-1} + 2k_3 c_{3,m}(c_{6\mathrm{tot},m} - c_{6,m}) - k_5 c_{6,m}$
extracellular acetaldehyde/pyruvate	$\frac{dc_{4,\mathrm{ex}}}{dt} = -r_7 + \frac{\varphi}{N_C}\sum_{i=1}^{N_C} J_m = -kc_{4,\mathrm{ex}} + \frac{\varphi}{N_C}\sum_{m=1}^{N_C}\kappa(c_{4,m} - c_{4,\mathrm{ex}})$

[a] For the sake of clarity, the notation of the metabolic model employed by Wolf and Heinrich[24] and Henson et al.[6] has been adapted to be consistent with that used in this work (cf Scheme 1): i.e., our $c_{k,m}$ equals their $S_{k,i}$ ($k = 1-4$), and $r_{j,m} = v_{j,i}$ except for $c_{5,m} = N_{2,i}$, $c_{6,m} = A_{3,i}$, $c_{5\mathrm{tot}} = N$, $c_{6\mathrm{tot}} = A$, and $N_C = M$. To allow for a shortened notation, all concentrations denoting $c_{i,m,\mathrm{in}}$ have been abbreviated as $c_{i,m}$.

Table 2. Nominal Parameter Values[a] of the Yeast-Cell-Ensemble Model (after Wolf and Heinrich[24] and Henson et al.[6])

parameter	value	parameter	value
J_0	2.30 mM·min^{-1}	κ	50.0 min^{-1}
k_1	100 mM^{-1}·min^{-1}	Q	4.00
k_2	6.00 mM^{-1}·min^{-1}	K_I	0.520 mM
k_3	16.0 mM^{-1}·min^{-1}	$c_{5\mathrm{tot},m}$[b]	1.00 mM
k_4	100 min^{-1}	$c_{6\mathrm{tot},m}$[b]	4.00 mM
k_5	1.28 min^{-1}	φ	0.1
k_6	12.0 mM^{-1}·min^{-1}	N_C	100 000
k	1.30 min^{-1}		

[a] All values have been taken from Henson et al.,[6] except for N_C (this study). [b] In this work $c_{5\mathrm{tot},m}$ and $c_{6\mathrm{tot},m}$ correspond to the parameters N and A, respectively, in Wolf and Heinrich[24] and Henson et al.[6].

Table 3. Initial Values for a Point on the Oscillatory Solution Corresponding to the Parameter Values in Table 2 as Given by Henson et al.[6]

variable[a]	value	variable[a]	value
$c_{1,m}$	1.023 mM	$c_{5,m}$	1.624×10^{-1} mM
$c_{2,m}$	6.696×10^{-1} mM	$c_{6,m}$	1.250 mM
$c_{3,m}$	7.194×10^{-2} mM	$c_{4,\mathrm{ex}}$	8.419×10^{-2} mM
$c_{4,m}$	1.094×10^{-1} mM		

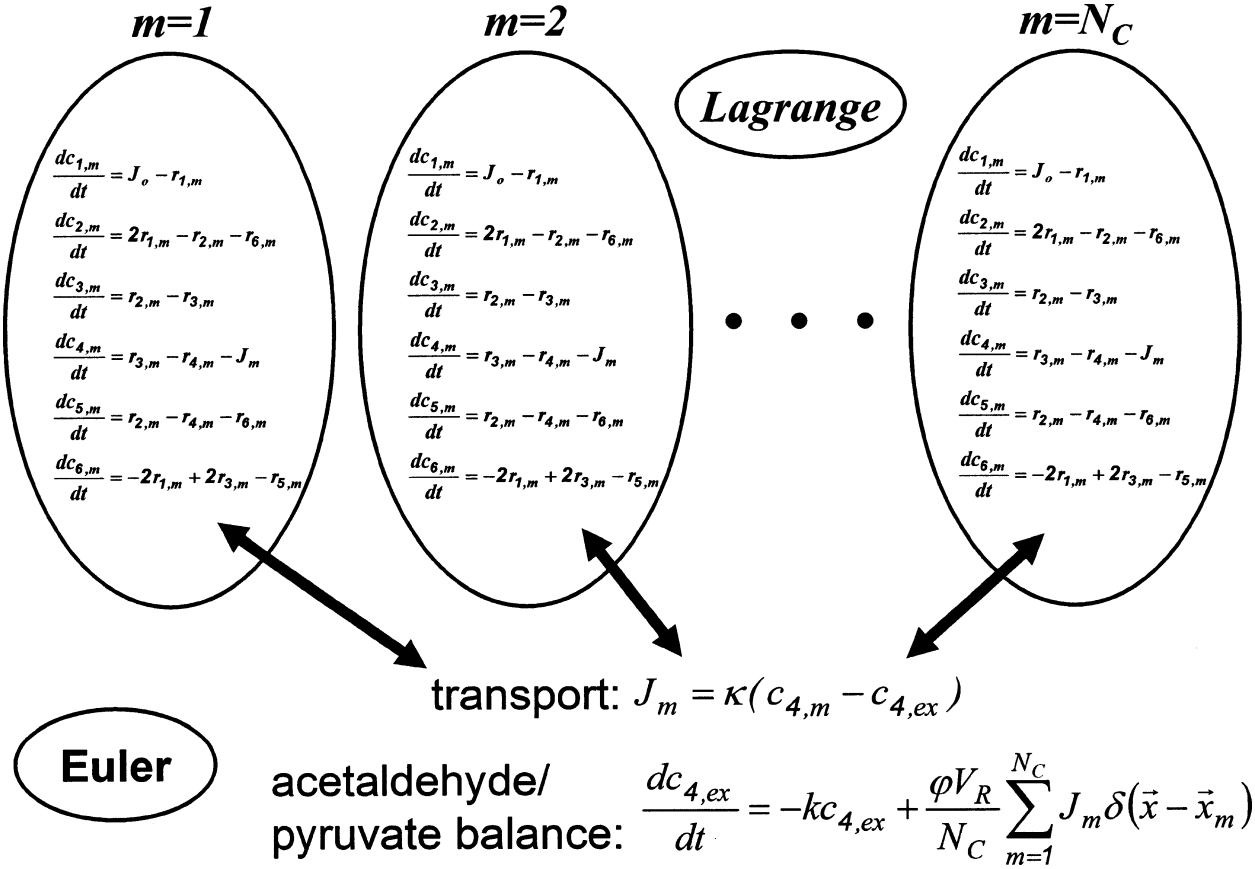
[a] For meanings of the variable names, refer to Table 1.

intracellular concentrations from their nominal values (cf Table 3) according to a Gaussian distribution with zero mean and a variance of 2.25 in Henson et al.,[6] whereas a uniform distribution between zero and twice the nominal value is employed in this work. The intracellular level of the coenzyme NADH serves as a measure to assess the degree of synchronization between the different cells. Figure 5 displays the temporal evolution of the intracellular NADH concentration of 1000 cells for the ideally mixed case without any extracellular concentration gradients. The initially unsynchronized population of cells exhibits increasing synchronization with time because of the coupling effect of extracellular acetaldehyde/pyruvate until all cells oscillate with equal amplitude and phase after about 60 min.

In the corresponding simulations with the Euler–Lagrange approach for 100 000 cells, the transport phenomena in the fluid phase are superimposed, thus linking the metabolically structured and segregated model for the dynamic behavior of the biophase with the three-dimensional turbulent flow in the bioreactor. Two distinct settings have been analyzed for the 68-L laboratory-scale fermentor described above:[23] (1) turbulent flow ($N_{\mathrm{imp},1} = 165$ rpm) and (2) low-turbulence conditions ($N_{\mathrm{imp},2} = 55$ rpm). When starting with a desynchronized population (Figure 6a), the combined model predicts almost complete synchronization of the intracellular NADH dynamics in case 1 (Figure 6b), although it occurs more slowly than in the ideally mixed reactor described above (cf Figure 5). Also, the final degree of synchrony is slightly diminished under conditions of nonideal mixing as compared to the ideally mixed case. However, in the instance of slow agitation, where turbulent dispersion is reduced by decreasing the impeller speed from $N_{\mathrm{imp},1} = 165$ rpm to $N_{\mathrm{imp},2} = 55$ rpm after synchronization has been achieved, significant concentration gradients within the extracellular space occur. This leads to a dramatic loss of synchrony (cf Figure 6c) and eventually to almost complete desynchronization. Figure 7 displays snapshots of the combined spatial and temporal dynamics of the intracellular NADH of the cell population in the reactor for both conditions described above. This result demonstrates that the extracellular coupling mechanism offers only limited robustness to variations in the intensity of mixing.

It should be stressed that the example of yeast glycolytic oscillations has been chosen solely to illustrate the potential of the simulation approach presented. The authors are well aware that this type of oscillation has no relevance in practical bioprocesses. However, we aim

Scheme 2. Extracellular Coupling in a Population of N_C Yeast Cells[a]



[a] The dynamic state of the individual cell is represented by a set of six dynamic balance equations. Coupling between the members of the cell population ensues because of uptake or excretion of the extracellular attractor acetaldehyde/pyruvate (c_4).

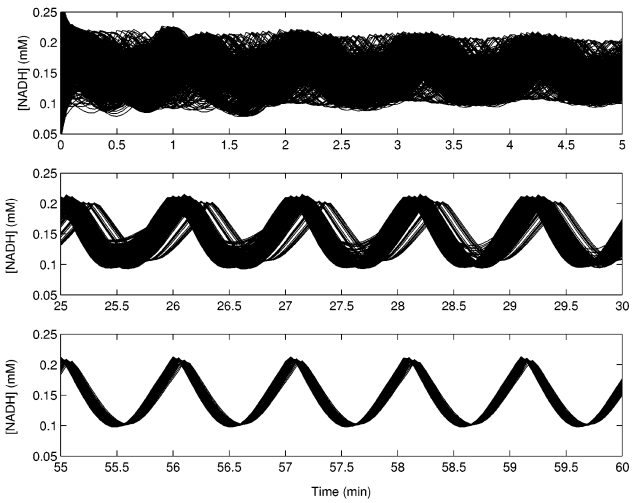


Figure 5. Synchronization of the autonomous oscillations of yeast cells at the population level. [Reproduced with permission from Henson et al. *Biochem. J.* **2002**, *368* (Pt.2), 443−446, © The Biochemical Society.]

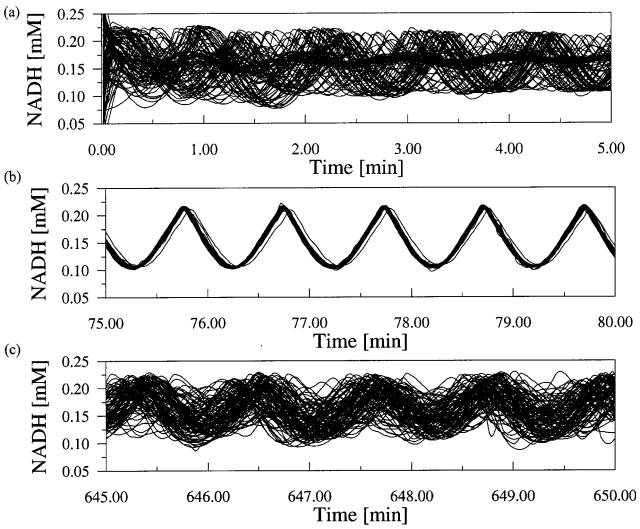


Figure 6. Synchronization of autonomous oscillations in a yeast cell population under nonideal mixing conditions simulated using the Euler−Lagrange approach for the vessel described in Figure 2. The time course of intracellular NADH is depicted for 100 of the simulated 100 000 cells. (a) Disorganized oscillations 5 min after startup with a desynchronized population at a stirrer speed of $N_{imp,1} = 165$ rpm. (b) Synchronized oscillations after 70 min for the same setting. (c) Same simulation as in part b, but after switching to a stirrer speed of $N_{imp,2} = 55$ rpm at $t = 80$ min; a dramatic loss in the synchrony of the oscillations is observed after 600 min.

to apply the cell-ensemble approach to the closely related but practically more relevant case of cell-cycle oscillations observed in glucose-limited continuous cultures of *Saccharomyces cerevisiae*[25−27] in the future. Here, an intricate link exists between the cell-cycle position of the single cell and the regulation of sugar metabolism[27−31] that can profoundly affect the local substrate consumption and production rates in the reactor. Although these oscillations are different in period and biochemical mechanism, they can formally be described by the same type of model because they are also presumed to be mediated via exchange of an extracellular attractor.[32,33] Therefore, the stability of

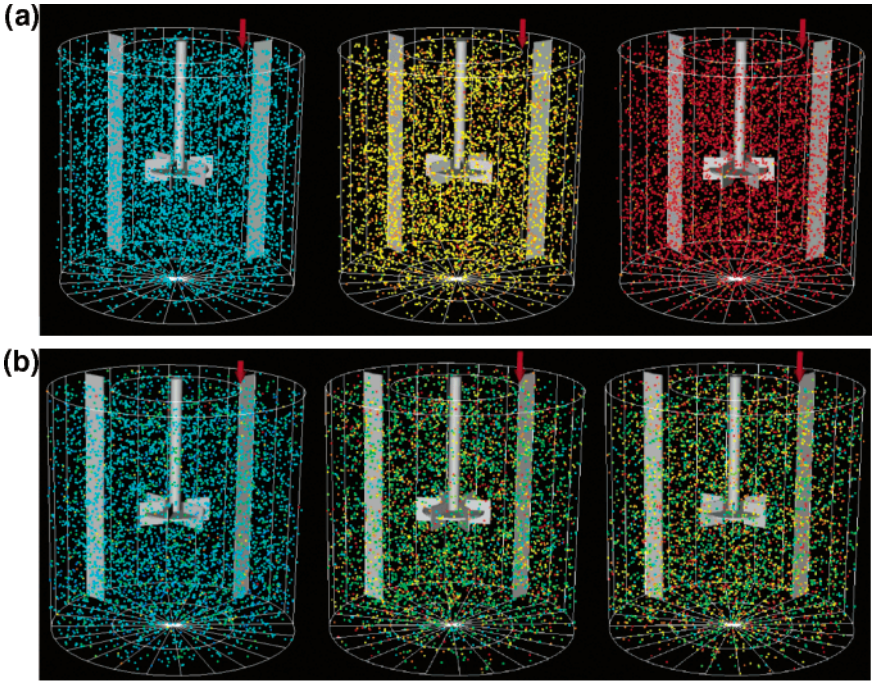


Figure 7. Movement of a population of 100 000 oscillating yeast cells traveling through the three-dimensional turbulent flow field of a stirred-tank bioreactor (same vessel as in Figure 2). (a) Synchronization under well-mixed conditions ($N_{imp,1}$ = 165 rpm), (b) desynchronization of the autonomous oscillations due to concentration gradients of the attractor acetaldehyde/pyruvate at lower agitation speed ($N_{imp,2}$ = 55 rpm). The prismatic colors of the particles indicate the intracellular NADH concentrations, which ascend from magenta (0.1 mM) via cyan, green, and yellow to red (0.22 mM).

these cell-cycle oscillations is expected to exhibit the same sensitivity to mixing effects as the purely glycolytic oscillations simulated here.

The results shown above for glycolytic oscillations are hence in good agreement with the fact that stable cell-cycle oscillations in yeast populations have so far mainly been observed in well-stirred laboratory-scale fermentors where almost perfect mixing can be assumed, but not in reactors that exhibit significant heterogeneity and thus concentration gradients, such as industrial-scale bioreactors.[32,34,35]

Conclusion

In the present work, an Euler−Lagrange approach has been applied to the description of the behavior of a heterogeneous cell population in a stirred-tank reactor. This method allows the population behavior to be described as the outcome of the interaction between the intracellular state of its individual cells and the turbulent flow field in the bioreactor. The chosen Euler−Lagrange representation of the cell-ensemble approach permits analysis of the lifelines of individual cells in space and time, as illustrated for the synchronization of autonomous glycolytic oscillations in yeast cells at the population level. This example represents the first integrated model combining CFD with a segregated description of a cell population in a stirred tank that simultaneously accounts for a detailed intracellular structure of the single cells. Even for the small reactor used in this example, the system of coupled nonlinear ODEs resulting after spatial discretization consists of about 1.3 million equations. Therefore, this problem can be solved only through the application of efficient numerical procedures. Despite the steady increase in the available computing power and the continuous development of more sophisticated solution strategies, it is even now impossible to solve such models using alternative representations, e.g., employing the well-known population balances. A numerically efficient, stable, and accurate solution of the underlying system of equations has been achieved employing a fractional-step method for system decomposition and a TVD approach for the spatial discretization. Problems of greater practical relevance that can be tackled in the future using the Euler−Lagrange description include the impact of sugar transport into bacterial cells containing a phosphotransferase system (PTS) or cumulative starvation effects in individual cells passing through substrate-depleted zones in a fermentor. Also, yeast cell-cycle oscillations as well as cell-cycle-specific product formation in yeast[36] or animal[37] cell cultures and segregated modeling of plasmid loss in populations of recombinant organisms[10,38] can be addressed. The presented Euler−Lagrange approach thus provides a potent simulation strategy for the design and operation of bioreactors, especially when the fate of the individual cell becomes the focus of interest. Furthermore, the modeling and numerical strategies presented in this paper should also prove to be useful in fields outside biochemical engineering where a detailed description of the corpuscular phase is required.

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